POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

F1 Real-Time Strategy

Tesi di Laurea Magistrale in
Computer Science and Engineering - Ingegneria Informatica

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Academic Year: 2025-26

Abstract

Keywords:

Abstract in lingua italiana

Parole chiave:

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1 Introduction

1.1. Context and Motivation

Template intent. Explain why Formula 1 pit-wall decision support is a strong case for streaming analytics plus machine learning.

- Start from the operational problem: decisions are sequential, time-critical, and made under uncertainty.
- Explain why batch-only analysis is insufficient for race-time decisions.
- Position event-time processing and low-latency architecture as enabling conditions.

Suggested references: [1, 8, 20, 31].

1.2. Research Scope and Objectives

Template intent. Define exactly what the thesis will and will not do.

- Scope: pit-stop recommendation as a supervised lap-level decision task in a streaming architecture.
- Comparison target: deterministic heuristic strategy engine vs learned model on the same replay stream.
- Explicit non-goals: full race-winning simulation, proprietary telemetry replication, and team-specific private strategy rules.
- Add 2–4 research questions, each measurable in Chapter 6.

Suggested references: [13, 19, 28, 29].

1.3. Contributions

Template intent. List the contribution claims in a thesis-safe and testable format.

- A reproducible event-time data engineering pipeline for F1 replay analytics.
- A leakage-aware ML protocol for pit-stop recommendation.
- A fair comparison protocol between heuristic and ML decisions in identical operating conditions.
- A reproducibility package (data contracts, scripts, environment, and experiment documentation).

Suggested references: [9, 23, 25].

1.4. Thesis Organization

Template intent. Provide a concise chapter roadmap.

- Chapter 2: state of the art and research gap.
- Chapter 3: problem definition, data, labels, and evaluation metrics.
- Chapter 4: solution design and comparison protocol.
- Chapter 5: implementation choices and reproducibility.
- Chapter 6: evaluation campaign and discussion.
- Chapter 7: conclusions, limitations, and future work.

2 State of the Art

2.1. Streaming Data Analytics

Template intent. Summarize stream-processing principles relevant to this thesis.

- Define unbounded data, event time, out-of-order events, watermarks, and correctness vs latency trade-off.
- Explain why a replayed historical race can still be processed as an event stream.
- Connect system guarantees to downstream ML reliability and reproducibility.

Suggested references: [1, 8, 20].

2.2. Complex Event Processing

Template intent. Explain CEP as the bridge between raw streams and tactical race signals.

- Define patterns, windows, sequence constraints, and event-time matching.
- Motivate CEP for lift-and-coast, tire-drop, and other race-state transitions.
- Clarify how CEP outputs differ from ML outputs and why both are needed in your architecture.

Suggested references: [1, 8, 31].

2.3. Platforms for Stream and Event Processing

2.3.1. Apache Kafka

Template intent. Describe Kafka role in your system.

- Topic partitioning, ordering assumptions, replayability, and decoupling of producers/consumers.

- Why replay-by-offset is essential for deterministic audits and ML dataset regeneration.

Suggested references: [20].

2.3.2. Apache Flink

Template intent. Describe Flink role in your system.

- Event-time operators, stateful processing, checkpointing, and fault tolerance.
- Why Flink is suitable for both online alerting and offline reproducible extraction.

Suggested references: [1, 8].

2.3.3. Related Technologies and Trade-offs

Template intent. Briefly compare alternatives and justify your stack.

- Compare latency and semantics trade-offs at a high level.
- State why your selected stack best matches thesis goals: correctness, replayability, and integration simplicity.

Suggested references: [31].

2.4. Machine Learning for Motorsport and Strategy Optimization

Template intent. Organize literature by paradigm, then by methodology quality.

- Supervised classification paradigm for pit-stop timing.
- Optimization/planning paradigm for tire and stop sequencing.
- Reinforcement learning paradigm for sequential control.
- Explicitly separate domain results from methodology quality (splits, metrics, calibration, significance).

Suggested references: [9, 13, 19, 23, 25, 28, 29].

Methodology anchors to cite in this section: [2–6, 10–12, 14–17, 21, 22, 24, 26, 27, 30].

2.5. Applications in Formula 1 Analytics

Template intent. Focus on direct F1 evidence and identify operational gaps.

- What has been done with public F1 data (features, targets, models, evaluation).
- What is still missing (streaming deployment, fair heuristic-vs-ML comparison, robust validation design).
- Position tire degradation modeling as supporting signal for pit-stop decisions.

Suggested references: [7, 13, 18, 19, 28].

2.6. Summary and Research Gaps

Template intent. Close Chapter 2 with a short, explicit gap list that leads into Chapter 3.

- Gap 1: Most published F1 ML evidence is batch-oriented.
- Gap 2: Validation and leakage controls are often weak or under-reported.
- Gap 3: Probability quality and statistical significance are rarely central.
- Gap 4: Direct heuristic-vs-ML comparisons in a streaming setting are limited.

3 | Problem Statement and Data

3.1. Formula 1 Pit Stop Decision Problem

Template intent. Define the prediction task unambiguously.

- Inference unit: driver-lap decision point.
- Target: binary label, pit in next K laps (state your fixed K and rationale).
- Operational interpretation: decision support signal, not autonomous control.

Suggested references: [9, 13, 19, 28, 29].

3.2. Data Sources and Acquisition

3.2.1. FastF1 Data Interface

Template intent. Explain source provenance and reproducibility constraints.

- Seasons/races included, extraction process, and versioning policy.
- Data quality caveats (missing telemetry, weather noise, incidents).

Suggested references: [13, 19, 28].

3.2.2. Streaming Inputs and Event Types

Template intent. Define each stream as a contract.

- Telemetry stream, lap stream, and track-status stream.
- Event-time semantics, ordering assumptions, and replay settings.

Suggested references: [1, 8, 20].

3.3. Dataset Description

3.3.1. Telemetry Stream

Template intent. List fields and their modeling role in few words.

- Mention sampling rate, key signals, and normalization choices.
- State known limitations and filtering rules.

3.3.2. Lap Stream

Template intent. Describe lap-level context and pace variables.

- Timing, stint, compound, tire age, and gap-derived features.
- Explain why lap-level aggregation is aligned with pit decision timing.

3.3.3. Track Status Stream

Template intent. Describe race-neutralization context.

- Encoding of green/yellow/VSC/SC and time availability constraints.
- How this stream is used for filtering, labeling, and stress tests.

3.4. Target Definition and Labeling Strategy

Template intent. Make labeling auditable and leakage-safe.

- Exact positive/negative label rule and boundary conditions.
- Inclusion/exclusion policy (for example wet phases, early laps, unresolved events).
- Explicit anti-leakage policy: no derived heuristic outputs as ML inputs.

Suggested references: [6, 19, 26, 28].

3.5. Evaluation Metrics

Template intent. Define metrics before results to avoid metric shopping.

- Primary discrimination metrics for imbalance: precision, recall, F1, AUC-PR.
- Calibration metrics: Brier score and reliability analysis.

- Statistical comparison protocol: paired significance tests with corrected interpretation.

Suggested references: [5, 10–12, 17, 21, 22, 24, 27, 30].

4 | Solution and Design

4.1. System Architecture Overview

Template intent. Describe your end-to-end architecture in one page.

- Producer, broker, stream processor, data lake, and dashboard roles.
- Explain how architecture choices support reproducibility and fair comparisons.

Suggested references: [1, 8, 20].

4.2. Dataflow and Processing Topology

Template intent. Specify operator order and data dependencies.

- Event-time path from ingestion to features/alerts/labels.
- Stateful operators and where context enters the pipeline.
- Contracts for sinks used by audit and ML.

Suggested references: [8, 31].

4.3. Heuristic Strategy Engine Design

Template intent. Explain the handcrafted strategy logic at a conceptual level.

- Inputs, decision factors, and prioritization logic.
- Expected strengths and known failure modes.
- Clarify this module as baseline and operational comparator.

Suggested references: [9, 25].

4.4. Machine Learning Formulation

Template intent. Specify model-family choices and data protocol without reporting outcomes yet.

- Candidate baselines (for example logistic/XGBoost and optional sequence model).
- Feature groups, leakage controls, and temporal/grouped split strategy.
- Calibration and thresholding plan for decision outputs.

Suggested references: [6, 13, 17, 19, 21, 24, 26, 28].

4.5. Comparison Protocol, Heuristic vs ML

Template intent. Define a fairness protocol before experiments.

- Same replay windows, same data availability assumptions, same excluded contexts.
- Report both predictive quality and operational behavior.
- Predefine significance testing and uncertainty reporting.

Suggested references: [11, 12, 22, 30].

5 | Implementation Experience

5.1. Engineering Stack and Environment

Template intent. Keep this section factual and reproducible.

- Report runtime stack, versions, infrastructure, and packaging.
- Document environment constraints and reproducibility notes.
- Describe what is deterministic vs what can vary between runs.

5.2. Pipeline Implementation

5.2.1. Ingestion and Replay Components

Template intent. Describe implementation decisions that affect data fidelity.

- Replay timing policy, topic schema, and serialization choices.
- Error handling and idempotency notes.

5.2.2. Flink Operators and Real-time Alerts

Template intent. Summarize operator responsibilities and state management.

- Per-operator responsibilities and key state variables.
- Watermarks/checkpointing and practical trade-offs encountered.

5.2.3. Data Lake and Audit Layer

Template intent. Explain output contracts and quality checks.

- Sink layout, partitioning logic, and schema consistency policy.
- Audit scripts and contract-validation process.

5.3. Model Training Pipeline

Template intent. Keep this broad until final experiments stabilize.

- Training data extraction path and feature assembly overview.
- Split strategy implementation and anti-leakage safeguards.
- Hyperparameter search, calibration step, and model artifact versioning.

Suggested references: [6, 17, 21, 24, 26].

5.4. Real-time Inference Integration

Template intent. Describe integration options and chosen approach without overcommitting to one architecture in early drafts.

- Integration pattern selected (in-engine, service, or decoupled consumer).
- Input contract, output topic/schema, and latency budget assumptions.
- Fallback and observability strategy for inference failures.

6 Evaluation Campaign

6.1. Experimental Setup and Parameters

Template intent. Define the evaluation setting before showing any numbers.

- Races/seasons included, replay speed policy, and compute environment.
- Fixed parameters and tunable parameters.
- Repetition policy for robustness.

6.2. Dataset Split and Validation Protocol

Template intent. Explain exactly how leakage is prevented.

- Grouped/temporal split strategy and rationale.
- What is held out, when, and why.
- Stress-test subsets (for example SC/VSC/wet) vs core training subset.

Suggested references: [6, 26].

6.3. Baseline Definitions

6.3.1. Heuristic Baseline

Template intent. Specify heuristic decision rule version used in the comparison.

- Inputs available at inference time.
- Decision output format and thresholding policy.

6.3.2. Machine Learning Baseline(s)

Template intent. Declare model variants and calibration/threshold steps.

- Model families and selected final candidates.
- Calibration method and operating-point selection logic.

Suggested references: [5, 10, 17, 21, 24, 27].

6.4. Results on Recommendation Quality

Template intent. Add final tables/figures only when experiments are frozen.

- Report precision, recall, F1, AUC-PR, and calibration metrics.
- Provide race-level and aggregate-level views.
- Include uncertainty intervals and significance tests.

Suggested references: [11, 12, 22, 30].

6.5. Latency and Throughput Results

Template intent. Keep this section focused on operational feasibility.

- End-to-end latency distribution and throughput under replay loads.
- Compare overhead introduced by ML inference path vs heuristic-only path.

6.6. Error Analysis and Failure Modes

Template intent. Explain where and why decisions fail.

- Analyze errors by race context (track, compound, neutralization regime).
- Distinguish data issues, model limitations, and unavoidable ambiguity.

6.7. Discussion

Template intent. Synthesize findings without overclaiming.

- Interpret practical implications for pit-wall support.
- State what can be concluded confidently and what remains uncertain.

7 Conclusion

7.1. Obtained Results

Template intent. Keep this section high-level until final numbers are fixed.

- Summarize what was built and evaluated.
- Restate outcome at the level of evidence (not hype): where ML improved, matched, or underperformed heuristic logic.
- Mention both decision quality and operational feasibility.

7.2. Limitations

Template intent. Be explicit and concrete.

- Data limitations (coverage, label ambiguity, special race conditions).
- Method limitations (model class, assumptions, calibration drift).
- System limitations (latency budget, integration constraints, observability).

7.3. Future Work

Template intent. Propose realistic next steps tied to thesis findings.

- Drift-aware adaptation and online recalibration.
- Richer decision spaces (for example compound recommendation jointly with timing).
- Tighter integration patterns and broader season-level validation.

Suggested references: [2–4, 15, 16].

7.4. Overall Discussion

Template intent. Close by positioning your thesis against the three strategy paradigms.

- Optimization/planning paradigm.
- Reinforcement-learning paradigm.
- Supervised classification in streaming deployment.

Suggested references: [9, 13, 19, 25, 28, 29].

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B.1. Configuration Parameters

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Variable	Description	Unit

Acknowledgements